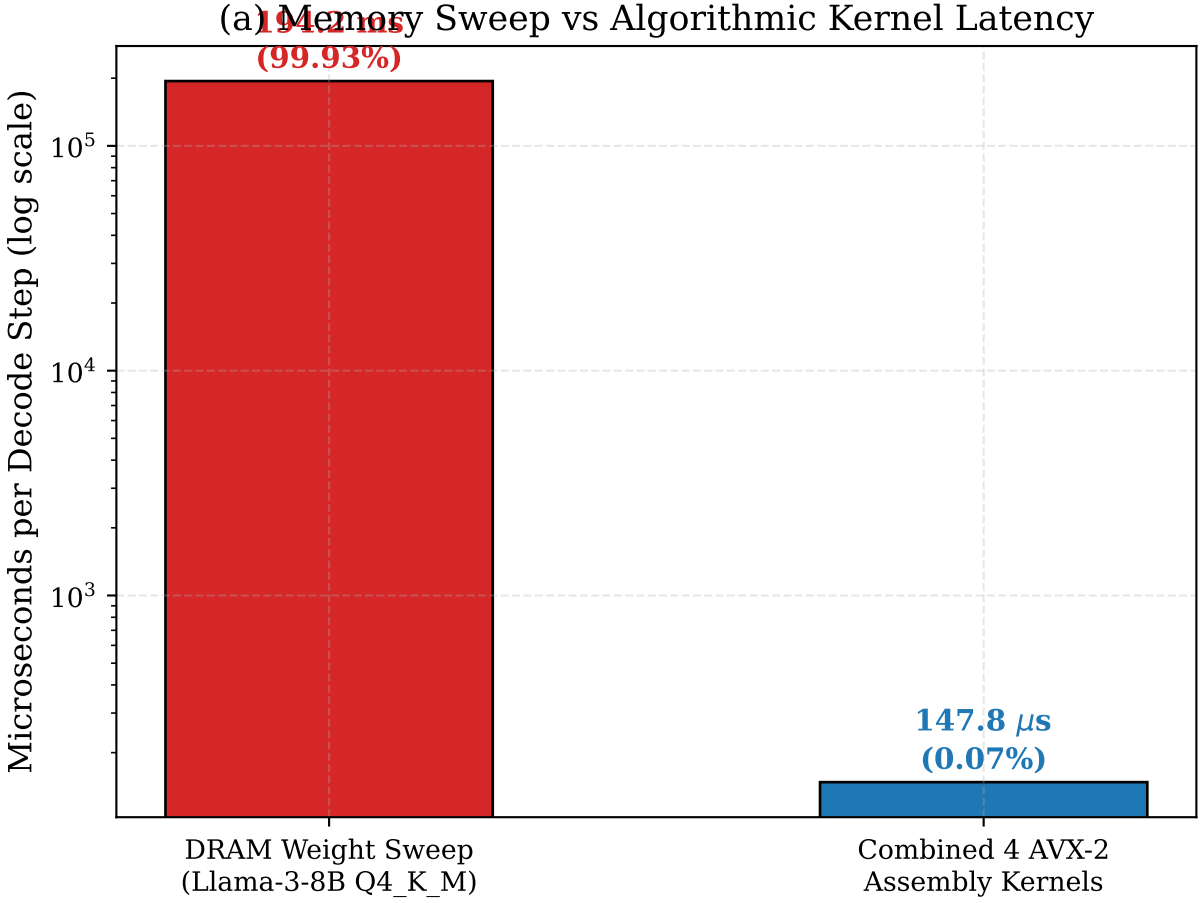
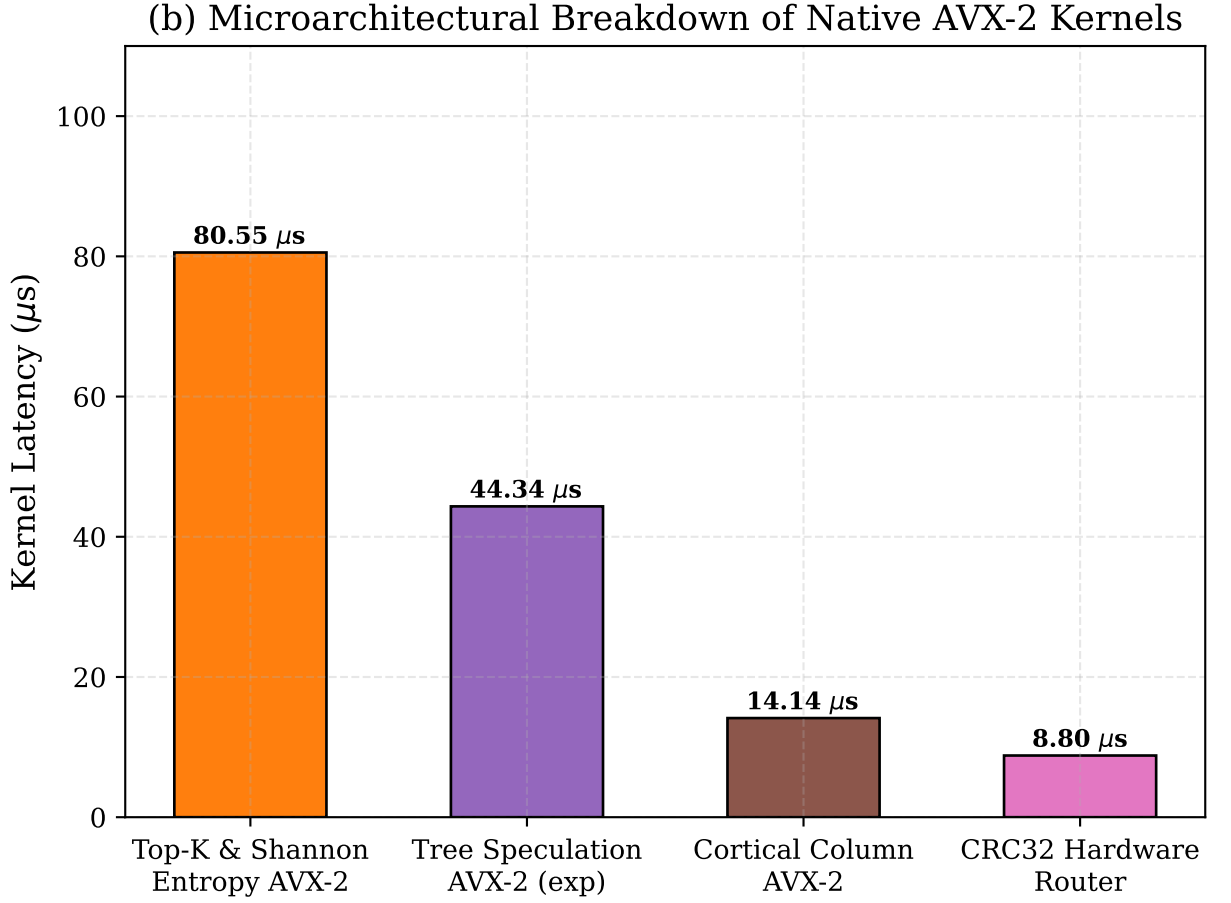

(a) Memory Sweep vs Algorithmic Kernel Latency
Microseconds per Decode Step (log scale)
10^5
10^4
10^3
194.2 ms
(99.93%)
147.8 μs
(0.07%)
DRAM Weight Sweep
(Llama-3-8B Q4_K_M)
Combined 4 AVX-2
Assembly Kernels


(b) Microarchitectural Breakdown of Native AVX-2 Kernels
Kernel Latency (μs)
100
80
60
40
20
0
80.55 μs
44.34 μs
14.14 μs
8.80 μs
Top-K & Shannon
Entropy AVX-2
Tree Speculation
AVX-2 (exp)
Cortical Column
AVX-2
CRC32 Hardware
Router